

Chapter 2. PERMUTATIONS

2.3 The Alternating group



Today we will collect some important properties of the alternating group A_n (the subset of S_n consisting on all even permutations).

Some important properties of A_n

Definitions 2.21. A permutation σ is called **even** or **odd** if it can be written as a product of an even or odd number of transpositions. The set of all even permutations in S_n is denoted A_n , and is called the **alternating group of degree n** .

THEOREM 2.23. *Let $n \geq 2$. Then:*

- (1) *The identity permutation ε is in A_n .*
- (2) *If σ and τ are in A_n , then σ^{-1} and $\sigma\tau$ are in A_n .*
- (3) $|A_n| = \frac{1}{2}n!$

Proof.

(1) follows from Lemma 2.22.

$$\varepsilon = (1\ 2)(1\ 2)$$

(2) Suppose now that $\sigma, \tau \in A_n$. Then we can write

$$\sigma = \tau_1 \tau_2 \dots \tau_r, \quad r \text{ even}, \quad \tau_i \text{'s transpositions}$$

$$\tau = \delta_1 \delta_2 \dots \delta_s, \quad s \text{ even}, \quad \delta_i \text{'s transpositions}$$

Then $\sigma\tau = \tau_1 \tau_2 \dots \tau_r \delta_1 \delta_2 \dots \delta_s$ is a product of $r+s$ transpositions. Since r and s are even, we have that $r+s$ is even, and by the Parity Theorem, we obtain that $\sigma\tau$ is even, that is, $\sigma\tau \in A_n$.

$$\underline{\sigma^{-1} \stackrel{?}{\in} A_n}.$$

$$\text{We first prove that } \sigma^{-1} \stackrel{?}{=} \underbrace{\tau_r \tau_{r-1} \dots \tau_2 \tau_1}_{\alpha}.$$

$$\underline{\sigma \alpha \stackrel{?}{=} e}$$

We check
this here

$$\underline{\alpha \sigma \stackrel{?}{=} e}$$

$$\begin{aligned}
 \sigma &= (\tau_1 \dots \tau_r)(\tau_r \tau_{r-1} \dots \tau_1) \quad \tau_r^2 = \varepsilon \\
 &= \tau_1 \dots \tau_{r-1} \tau_r^2 \tau_{r-1} \dots \tau_1 = \tau_{r-1}^2 = \varepsilon \\
 &= \tau_1 \dots \tau_{r-2} \tau_{r-1}^2 \tau_{r-2} \dots \tau_1 = \tau_{r-2}^2 = \varepsilon \\
 &\vdots \\
 &= \tau_1^2 = \varepsilon
 \end{aligned}$$

This shows that

$$\sigma^{-1} = \tau_r \tau_{r-1} \dots \tau_1,$$

which is a product of an even number of transpositions. Then by the Parity Theorem we obtain that

$$\underline{\sigma^{-1} \in A_n}.$$

$$(3) \quad |A_n| \stackrel{?}{=} \frac{1}{2} n!$$

Write O_n to denote the set of all odd permutations in S_n . Then we have

$$\left. \begin{array}{l} S_n = A_n \cup O_n \\ A_n \cap O_n = \emptyset \end{array} \right\} \Rightarrow \left[\begin{array}{l} n! = |S_n| = \\ = |A_n| + |O_n| \end{array} \right]$$

Goal : $|A_n| \stackrel{?}{=} |O_n|$

We need to find a bijection $f: A_n \rightarrow O_n$.
Let $\gamma = (1 \ 2)$ and define

$$f(\sigma) = \sigma\gamma \text{ for all } \sigma \in A_n$$

Notice that $\sigma\gamma \in O_n$ because σ is an even permutation (it can be

written as an even number of transpositions) and we are adding one more transposition, namely, γ to such decomposition.

f is injective: $f(\sigma_1) = f(\sigma_2) \stackrel{?}{\Rightarrow} \sigma_1 = \sigma_2$

$$f(\sigma_1) = f(\sigma_2) \Rightarrow \sigma_1 \gamma = \sigma_2 \gamma \Rightarrow \text{Multiplying by } \gamma \text{ on the right}$$

$$\Rightarrow \sigma_1 \gamma^2 = \sigma_2 \gamma^2 \xrightarrow{\gamma^2 = E} \underline{\sigma_1 = \sigma_2}$$

f is surjective: for $z \in D_n$,

consider $\alpha = z\gamma \in A_n$ and $\gamma^2 = E$

$$\underline{f(\alpha)} = \alpha \gamma = (z\gamma)\gamma = z\gamma^2 = \underline{z}$$

We will come back to this very soon! 😊

Remark 2.24. A set of permutations is called a **group of permutations** if it contains the identity permutation, the product of any two of its members, and the inverse of any member. Hence S_n is a group, and Theorem [2.23](#) tells us that A_n is a group too.